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De cartesische coördinaten van $\left(3, -\frac{5\pi}{6}\right)$; (x, y) –?

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \Rightarrow \begin{cases} x = 3 \cos \left(-\frac{5\pi}{6}\right) \\ y = 3 \sin \left(-\frac{5\pi}{6}\right) \end{cases}$$

$$\Rightarrow \left(-\frac{5\pi}{6}\right) \left(\frac{180^\circ}{\pi}\right) = 150^\circ \Rightarrow 180^\circ - 150^\circ = 30^\circ \Rightarrow \begin{cases} \cos 30^\circ = \frac{\sqrt{3}}{2} \\ \sin 30^\circ = \frac{1}{2} \end{cases}$$

$$\Rightarrow x = 3 \cos \left(-\frac{5\pi}{6}\right) = 3 \left(-\frac{\sqrt{3}}{2}\right) = -\frac{3\sqrt{3}}{2}$$

$$\Rightarrow y = 3 \sin \left(-\frac{5\pi}{6}\right) = 3 \left(-\frac{1}{2}\right) = -\frac{3}{2}$$

Resultaat: $\left(-\frac{3\sqrt{3}}{2}, -\frac{3}{2}\right)$

References